

KALLURI SHAREEF BABU

Assistant Professor, School of Electrical Engineering || Manipal Institute of Technology Bengaluru, MAHE Manipal, India
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EDUCATION

- **Ph.D.** || National Institute of Technology Karnataka (NITK) Surathkal, KAR, India
Thesis: [Automatic Estimation of Personal Characteristics using Speech Data](#)
Department of Electronics and Communication Engineering July 2021
Specialization in Machine Learning and Speech Signal Processing
Cumulative GPA: 7.68/10.0
- **M.Tech.** || Jawaharlal Nehru Technological University, Anantapur Anantapur , AP, India
Department of Electronics and Communication Engineering Jun 2013
Specialization in Digital Electronics and Communication Systems
Percentage: 82.44%
- **B.Tech.** || Jawaharlal Nehru Technological University, Anantapur Anantapur , AP, India
Department of Electronics and Communication Engineering Jun 2011
Percentage: 67.76%.

RESEARCH EXPERIENCE

- **Post-Doc Researcher** November 2023 - Nov 2024
[Learning and Extraction of Acoustic Patterns \(LEAP\) Lab](#),
Dept. of Electrical Engineering, Indian Institute of Science(IISc), Bangalore, India.
Projects Involved:
 - * Diarization of Speaker and Language in Conversational Environments ([DISPLACE](#))
[Lead organiser of DISPLACE challenge at Interspeech 2024](#)
 - * Speech Analytics and Conversational AI, funded by British TelecomSupervisor: [Prof. Sriram Ganapathy](#)
- **Post-Doc Researcher** May 2021 - March 2023
[Intelligent User Interface Lab](#),
Dept. of Computer and Information Science, Seikei University, Tokyo, Japan.
Projects Involved:
 - * User-adaptive artificial intelligence for human-computer interaction (PANORAMA)
project funded by JST, AIP Japanese-German-French AI Research program
 - * The Realization of an Avatar-Symbiotic Society where Everyone can Perform Active Roles without Constraint (Project manager is Professor Hiroshi Ishiguro at Osaka University) funded by JST Moonshot R&D goal 1: Realization of a society in which human beings can be free from limitations of body, brain, space, and time by 2050Supervisor: [Prof. Yukiko I Nakano](#)

- **Research Scholar** December 2013 to 2020
Department of Electronics and Communication Engineering,
National Institute of Technology Karnataka Surathkal.
Supervisor: [Dr. Deepu Vijayasenana](#)
- **Project Research Assistant (SERB Funded)** January 2019 to October 2020
Department of Electronics and Communication Engineering,
National Institute of Technology Karnataka Surathkal.
Title: Automatic Multilingual Speaker Profiling and Forensics.
Supervisor: [Dr. Deepu Vijayasenana](#)

Visiting Research Scholar: Learning and Extraction of Acoustic Patterns ([Leap Lab](#)),
Department of Electrical Engineering, Indian Institute of Sciences (IISc) Bangalore.
- **Research Intern** October 2012 to May 2013
National Atmospheric Research Laboratory ([NARL](#)), Gadanki, Tirupati, AP, India.
Topic: Implementation of Reed-Solomon Decoder Using Microblaze for Radiosonde Telemetry system..
Mentor: Dr. T.V.C. Sarma, Scientist/Engineer - SG

TEACHING EXPERIENCE & CURRENT PROJECTS

- **Assistant Professor** at Manipal Institute of Technology Bengaluru, MAHE Manipal, India
June 2026 - Till Date
- **Assistant Professor** at UPES, Dehradun
Subjects Taught Dec 2024 - June 2026
 - * Digital Electronics
 - * Database Engineering, DBMS, Java
- **Current Projects**
 - Organising for [DISPLACE-M](#), in collaboration with IISc, NITK and IIT Bombay.

PUBLICATIONS

- **Journal:**
 1. Kalluri, Shareef Babu, Deepu Vijayasenana, and Sriram Ganapathy, “Automatic speaker profiling from short duration speech data”, [Speech Communication](#) (2020). (SCIE Indexed, Impact factor: 3.2)
 2. Sugandha Sharma, Virender kadyan, Shareef Babu Kalluri, “Sensor-Enabled Automated Bioacoustic Systems for Precision Livestock Management and Wildlife Welfare Monitoring: A Comprehensive Review.” International Journal on Smart Sensing and Intelligent Systems. ” (*Accepted*)
 3. Sugandha Sharma, Virender kadyan, Shareef Babu Kalluri, “Deep Learning Based Acoustic Decision Support System for Automated Estrus Detection in Water Buffaloes”, Journal of Springer Nature Computer Science.(Under review)
 4. Panduranga Ravi Teja, Prasad, M. V.D, Shareef Babu Kalluri, Karthik Nasani , B. Raja Srinivasa Reddy , Chaitanya Kumar Marpu “RD-Aug: A Reaction–Diffusion-Inspired Synthetic Augmentation Framework for Potato Blight Stage Classification”, Journal of Computers and Electronics in Agriculture. (Under-review)

• **Conferences:**

1. Sugandha Sharma, Virender kadyan, Shareef Babu Kalluri, “Fusion-based Deep Learning Approach for Estrus Detection for Low-Resource Animal audio Using Augmentation Techniques” 14th International Conference on Intelligent Systems and Embedded Design (ISED-2026) (*Accepted*).
2. Dhanya E, Ankita Meena, Manas Nanivadekar, Noumida A, Victor Azad, Ashwini Nagaraj Shenoy, Pratik Roy Chowdhuri, Shobhit Banga, Vanshika Chhabra, Chitralekha Bhat, Kalluri, Shareef Babu, Srikanth Raj Chetupalli, Deepu Vijayasenan, Sriram Ganapathy, “Benchmarking Speech Systems for Frontline Health Conversations: The DISPLACE-M Challenge”, [arXiv preprint arXiv:2603.02813](#) (2026). (Interspeech 2026 - Accepted)
3. Shareef Babu Kalluri, Parth Joshi, Anu Yadav “Deep Learning-Based Detection of Stroke-Induced Dysarthria Using Audio Spectrogram Representations,” in Second International Conference on Advances in Computer Science, Electrical, Electronics, and Communication Technologies (CE2CT-2026), Bhimtal, Nainital, Uttarakhand, India. July 2-4, 2026.
4. Kalluri, Shareef Babu , Deepu Vijayasenan, “Analysis of Speech Features in Identifying Client’s Change Talk in Motivational Interviewing,” in *38th IEEE Region 10 Conference 2025 TENCON 2025*, Kota Kinabalu, Malaysia.
5. Puneet Bawa, Virender Kadyan, Kalluri, Shareef Babu , “Autoencoder-Driven Latent Representation Learning for Language-Agnostic Disordered Speech Classification using a Universal Feature set”, 17th Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC), IEEE, 2025, Singapore.
6. Kalluri, Shareef Babu , Prachi Singh, Pratik Roy Chowdhuri, Apoorva Kulkarni, Shikha Baghel, Pradyoth Hegde, Swapnil sontakke, Deepak K T, Mahadeva Prasanna S R, Deepu Vijayasenan, Sriram Ganapathy, “The Second DISPLACE Challenge : DIarization of SPEaker and LAnguage in Conversational Environments,” in the proceedings of [Interspeech 2024](#), Kos, Greece 2024.
7. Kalluri, Shareef Babu , Deepu Vijayasenan, Sriram Ganapathy, Ragesh Rajan M, Prashant Krishnan, “NISIP: A Multi-lingual Multi-accent Dataset for Speaker Profiling”, in the proceedings of the *46th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. Toronto, Ontario, Canada, IEEE, 2021.
8. Kalluri, Shareef Babu , Deepu Vijayasenan, and Sriram Ganapathy, “ A Deep Neural Network Based End to End Model for Joint Height and Age Estimation from Short Duration Speech”, *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, IEEE, 2019, Brighton, UK.
9. Kalluri, Shareef Babu, Ashwin Vijayakumar, Deepu Vijayasenan, and Rita Singh, “ Estimating Multiple Physical Parameters from Speech data”, in *IEEE 26th International Workshop on Machine Learning for Signal Processing (MLSP)*, 2016, Salerno, Italy.
10. Kalluri, Shareef Babu and Deepu Vijayasenan, “ Robust features for automatic estimation of physical parameters from speech”, in *IEEE Region 10 Conference, TENCON 2017*, Penang, Malaysia.
11. Kotra Venkata Sai Ritwik, Kalluri, Shareef Babu, Deepu Vijayasenan, “ COVID-19 Detection from Spectral features on the DiCOVA Dataset”, in the proceedings of [Interspeech 2021](#), Brno, Czech Republic.
12. Deepu Vijayasenan, Kalluri, Shareef Babu, K Sreekanth, and Ansal Issac, “ Study of Wireless Channel Effects on Audio Forensics”, in *IEEE 22nd Annual International Con-*

ference on Advanced Computing and Communication (ADCOM), 2016, Bangalore, India.

- **Other Publications**

1. Kotra Venkata Sai Ritwik, Kalluri, Shareef Babu, Deepu Vijayaseenan, “ COVID -19 Patient Detection from Telephone Quality Speech Data”, [arXiv preprint arXiv:2011.04299 \(2020\)](#).

TEACHING ASSISTANTSHIP & MENTORING

- Digital Signal Processing, Digital Electronics, Micro processors and Embedded Systems (Lab)
- Pattern Recognition and Machine Learning (Theory)
- Mentored Interns during PhD and Post-Doc tenure

ASSISTED FOR PROJECT FUNDING

- Automatic Multilingual Speaker Profiling and Forensics (co-ordinating in grant fund release and technical report writing), funded by SERB, Govt of India.
- Speech Analytics and Conversational AI (Proposal, Technical report writing, Action plan for 2024-2025 year), funded by British Telecom
- National Language Translation Mission (NLTM) BHASHINI : IISc - NITK consortium (Technical report writing, and co-ordinating in grant fund release) funded by MEiTY, Govt of India.

FACULTY DEVELOPMENT PROGRAMMES (FDPS) ORGANISED & SESSION CHAIR

- Organizer, FDP on “Emerging Frontiers in Speech and Language Technologies”, UPES Dehradun & EICT Academy, IIITDM Jabalpur (MeitY, Govt. of India), Online, Jan 6–13, 2026.
- Session Chair, 17th IEEE International Conference on Computational Intelligence and Communication Networks (CICN 2025), NIT Goa, India, Dec 20–21, 2025.

FACULTY DEVELOPMENT PROGRAMMES (FDPS) ATTENDED

- Advances in Artificial Intelligence for Image Processing and Medical Applications, organised by UPES from March 3–7, 2025.
- Intelligent Systems and Emerging Technologies in Computing and Electronics, organised by UPES from July 21–25, 2025.
- Effective Pedagogy (Practice and Policy Alignment), organised by UPES from July 28–August 1, 2025.

INVITED TALKS AND ORAL PRESENTATIONS

- Oral presentation at the 38th IEEE Region 10 Conference (TENCON 2025), Kota Kinabalu, Sabah, Malaysia.
- Invited talk on “Diarization of Speaker and Language in Conversational Environments (DISPLACE)” at the EECS Symposium, Indian Institute of Science (IISc), Bangalore, April 2024.

POSTER AND PAPER PRESENTATIONS

- Poster presentation at Interspeech 2024, Kos Islands, Greece, September 2024.
- Poster presentation at ICASSP 2019, Brighton, United Kingdom, May 2019.
- Poster presentation at IEEE Region 10 TENCON 2017, Penang, Malaysia, November 5–8, 2017.
- Poster presentation at MLSP 2016, Salerno, Italy, September 2016.
- Paper presentation at ADCOM 2016, Bangalore, India, September 2016.

CONFERENCES, WINTER SCHOOLS, TRAININGS AND WORKSHOPS ATTENDED

- IBM certification Training on “ Machine Learning and Deep Learning”. Jan-2026
- INTERSPEECH 2018, Hyderabad, India, September 2018.
- Winter School on Speech and Audio Processing (WiSSAP 2020), Indian Institute of Technology Mandi, Himachal Pradesh, January 2020.
- Winter School on Speech and Audio Processing (WiSSAP 2019), College of Engineering, Trivandrum, Kerala, January 2019.
- Winter School on Speech and Audio Processing (WiSSAP 2015), Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar, Gujarat, January 2015.
- Seminar on “Fundamentals of Speech Analysis and Synthesis”, Department of Speech and Hearing, School of Allied Health Sciences, Manipal University, Manipal, India, October 2015.
- Workshop on “Fundamentals of Speech Processing”, Department of Computer Science and Engineering, NITK Surathkal, March 2014.
- National seminar on “Frontiers in Machine Learning and Speech Processing”, Department of Computer Science, Central University of Kerala, February 2015.

AWARDS & FELLOWSHIPS

- MHRD Fellowship for Ph.D., Government of India (December 2013 – December 2018).
- Awarded a grant of USD 280 for student volunteer coordination at ICASSP 2019, Brighton, United Kingdom, May 2019.
- IEEE Signal Processing Society Student Travel Grant (2016) for attending the 26th International Workshop on Machine Learning for Signal Processing (MLSP 2016), Salerno, Italy.
- TEQIP-II International Student Travel Grant (2016) for attending the 26th International Workshop on Machine Learning for Signal Processing (MLSP 2016), Salerno, Italy.
- First Prize, National Level Student Technical Symposium (INTERACT2K10), Anil Neerukonda Institute of Technology, Visakhapatnam, Andhra Pradesh, India (2010).

TOP SKILLS

- Programming Language: Python, Java, SQL
- Programming Tools (IDE): Matlab
- Python Libraries: Numpy, Scipy, Matplotlib, Pandas
- Research Interest: Speech Processing, Speaker diarization, Speech Pathology, Human Computer Interaction, Machine Learning, Deep Learning

PROFESSIONAL SERVICE & MEMBERSHIPS

- Reviewer, Journal of Speech Communication, Interspeech 2021 - 2026, IEEE INDICON 2025, ICASSP 2026.
- Reviewer, IEEE CONECCT 2020, SPCOM 2024, 2026, SPECOM 2023, Annual Conference on Intelligent User Interfaces (IUI) 2022 - 2024, CE2CT-2026, SCPGCON 2026
- Program Committee member – IMPACT-SPEECH 2026 (Co-located with EMNLP 2026)
- Member, ACM, ISCA, Senior Member – IEEE-SPS

[All the information written here is true to the best of my knowledge]

Kalluri Shareef Babu